Vincent Drenth

vdrenth@chicagobooth.edu | linkedin.com/in/vincentdrenth

Education

University of Chicago Booth School of Business — Chicago, US
PhD in Econometrics and Statistics — *Sep 2026 – present*

University of Oxford — Oxford, UK
MSc in Statistical Science — *Oct 2023 – Sep 2024*

- Mark: 78% (Distinction)
- Dissertation: Deep Bayesian Batch Active Learning for Classification with Kernels (85%)

Erasmus University Rotterdam — Rotterdam, NL
BSc in Econometrics and Operations Research, Minor in Computer Science — *Sep 2020 – Jul 2023*

- GPA: 9.2/10 – Cum Laude, Top 0.18% of 1,128 students
- Thesis: Nonlinear Supervised Dimension Reduction for Macroeconomic Forecasting

Research Experience

University of Oxford — Oxford, UK
MSc Dissertation: Deep Bayesian Batch Active Learning for Classification with Kernels — *Jun 2024 – Sep 2024*

- Developed a Bayesian active learning method for neural networks using Laplace-approximated posteriors and Fisher information, implementing log-determinant batch selection as a tractable proxy for Expected Information Gain.
- Demonstrated superior label-efficiency over BALD and entropy baselines on MNIST and a coarse ImageNet while remaining computationally tractable through greedy submodular optimization.
- Validated robustness on Repeated-MNIST (redundant data) and Dirty-MNIST (aleatoric uncertainty), outperforming baselines in scenarios with data redundancy and high noise.
- Supervisor: Prof. Tom Rainforth and dr. Andreas Kirsch

Erasmus School of Economics — Rotterdam, NL
Research Assistant — *Sep 2022 – Jun 2023*

- Designed and implemented simulation studies in R assessing a robust dimension-reduction method (via invariant coordinate selection) and downstream clustering for rating-scale data in the presence of outliers.
- Conducted literature reviews on survey design (measurement reliability, handling careless responding) and robust clustering methods.
- Helped prototype and implement a reactive R Shiny GUI for robust mediation analysis methods, contributing to the robmedExtra package (published on CRAN).
- Supervisor: Prof. Andreas Alfons.

Erasmus University Rotterdam — Rotterdam, NL
BSc Thesis: Nonlinear Supervised Dimension Reduction for Macroeconomic Forecasting — *Apr 2023 – Jul 2023*

- Developed a nonlinear forecasting pipeline combining distance-correlation screening, kernel PCA/autoencoders, and nonlinear regression, and evaluated its out-of-sample performance on FRED-MD using statistical forecast comparison tests.
- Supervisor: Dr. Daan Opschoor

Teaching Experience

Erasmus University Rotterdam — Rotterdam, NL
Teaching Assistant — *Sep 2022 – Jul 2023*

- Courses taught: Introduction to Analysis, Matrix Algebra, Analysis, Probability Theory, and Statistics.
- Led weekly tutorials, developing problem-solving strategies and explanations for theoretical concepts.

Work Experience

Turnleaf Analytics London, UK

Data Scientist *Sep 2024 – Jun 2026*

- Built daily inflation nowcasting models using mixed-frequency regression (MIDAS) to bridge high-frequency commodity and retail price data with monthly CPI, scaling to 100,000+ product-level time series.
- Adapted localized conformal prediction to improve coverage calibration across seasonal volatility patterns in inflation forecasting.
- Developed and validated macroeconomic forecasting models for Eurozone and US inflation, achieving a #3 ranking in Bloomberg's CPI surveys (50+ nowcasters).

Robeco Rotterdam, NL

Quantitative Research Intern *Jun 2023 – Aug 2023*

- Adapted and applied a deep conditional autoencoder framework to extract non-linear factors from a large set of corporate bond characteristics.
- Developed and implemented a differentiable rank-based learning objective to align model training with the economic ordering of asset returns.
- Supervisor: Dr. Philip Messow.

Technical Skills

Programming: Python (NumPy, SciPy, Numba, Cython), R (Tidyverse, data.table), SQL, MATLAB.

Frameworks: PyTorch, PyMC, scikit-learn, statsmodels.

Tools: LaTeX, Git, Linux, Bash, Docker.